

Lecture 04 — Inner product

Mon Aug 31

Last updated: August 31, 2026.

Reading: Boyd 1.4 (inner product); Boyd 1.1 stack and slice (Quiz 1 prep).

Learning objectives

By the end of class, students should be able to:

1. Compute a dot product.
2. Interpret it as a weighted sum.
3. Interpret its sign and magnitude as alignment.
4. Recognize orthogonal vectors from a zero dot product.

Fill in — stack and slice

A **stacked** (or **block**) vector lists the entries of one vector after another. If b is an m -vector and c is an n -vector, then

$$(b, c) = (b_1, \dots, b_m, c_1, \dots, c_n)$$

is an $(m + n)$ -vector. A scalar can be stacked with a vector: if a is a 3-vector, then $(1, a)$ is a _____-vector with first entry 1, then a_1, a_2, a_3 .

Stacking is **not** vector addition: (a, b) has length $m + n$, not m .

A **slice** (or **subvector**) takes a contiguous block of entries. Boyd's notation $x_{r:s}$ means the vector of entries $x_r, x_{r+1}, \dots, x_s$. It has length _____. The endpoints are **inclusive**.

Example: if $z = (1, -1, 2, 0)$, then $z_{2:3} =$ _____.

If a and b are 10-vectors and c is a 20-vector, then $(a, b, c_{3:13})$ has length _____.

Fill in

For n -vectors a and b , the **inner product** (or **dot product**) is the scalar

$$a^T b = a_1 b_1 + a_2 b_2 + \dots + a_n b_n = \sum_{i=1}^n \text{_____}.$$

It is only defined when a and b have the same _____.

If w is a weight vector and x is a data vector (same length), then $w^T x$ is a _____ of the entries of x , using the weights in w .

Geometrically in $\mathbb{R}^2$ (and similarly in $\mathbb{R}^3$): if $a^T b > 0$, the arrows point roughly the _____ direction; if $a^T b < 0$, they point roughly _____ directions; if $a^T b = 0$, the vectors are _____.

Two vectors with $a^T b = 0$ are **orthogonal** (perpendicular). The inner product is commutative: $a^T b =$ _____.

The quantity $a^T a = a_1^2 + \dots + a_n^2$ is always _____ (never negative).

Worked in class

W1. Let $b = (2, -1)$ and $c = (0, 3, 1)$. Write the stacked vector (b, c) explicitly. What is its length?

W2. Let $z = (1, -1, 2, 0)$. Write $z_{2:3}$ and $z_{1:2}$. If $x = (63, 59, 61, 66)$ is four daily high temperatures ($^{\circ}\text{F}$), write the 3-vector of day-to-day **changes** as $x_{2:4} - x_{1:3}$.

W3. Compute $u^{\top}v$ for $u = (1, 1)$ and $v = (1, -1)$. Sketch both arrows in $\mathbb{R}^2$ from the origin. How does the zero dot product match the picture?

W4. Three quiz parts have scores (out of 1):

$$g = (0.80, 0.90, 0.70),$$

with weights

$$w = (0.50, 0.30, 0.20).$$

Compute the weighted total $w^\top g$. (Same idea as a weighted average when the weights sum to 1.)

W5. Morning wind displacements (east, north), in km/h:

$$a = (3, 1), \quad b = (-1, 3).$$

Compute $a^\top b$. Is it positive, negative, or zero? In one sentence, what does the sign tell you about how the two directions line up?

Your turn

Y1. Let $a = (4, 1, 2)$ and $b = (1, 0, 1)$ be shopping lists (apples, bread, milk). Write (a, b) and state its length.

Y2. Let $c = (10, 12, 11, 14, 13)$ be five daily counts. Find $c_{2:4}$.

Y3. Compute $(2, -1)^\top(3, 4)$ and $(1, 0)^\top(0, 1)$.

Y4. A café tracks daily sales in the order (espresso, milk, syrup):

$$x = (40, 30, 10),$$

with prices in dollars

$$p = (3, 2, 1).$$

What quantity does $p^\top x$ represent? Compute it.

Y5. Two shopping lists, in the order (apples, bread, milk):

$$r = (2, 1, 3), \quad s = (4, 2, 6).$$

Compute $r^\top s$. Notice anything about r and s ? If $t = (1, 0, -1)$, compute $r^\top t$ and say whether r and t are orthogonal.

Answers (Your turn)

Y1. $(a, b) = (4, 1, 2, 1, 0, 1)$; length 6.

Y2. $c_{2:4} = (12, 11, 14)$.

Y3. $(2, -1)^T(3, 4) = 2 \cdot 3 + (-1) \cdot 4 = 2$. $(1, 0)^T(0, 1) = 0$ (orthogonal).

Y4. Total revenue in dollars: $p^T x = 3 \cdot 40 + 2 \cdot 30 + 1 \cdot 10 = 120 + 60 + 10 = 190$ dollars.

Y5. $r^T s = 2 \cdot 4 + 1 \cdot 2 + 3 \cdot 6 = 8 + 2 + 18 = 28$. Also $s = 2r$, so the lists are proportional (same “direction” in data space). $r^T t = 2 \cdot 1 + 1 \cdot 0 + 3 \cdot (-1) = 2 - 3 = -1 \neq 0$, so r and t are not orthogonal.